Machine Learning (ML)

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Types of Machine Learning

There are three main types of Machine Learning:

A. Supervised Learning

Definition: Supervised learning uses **labeled data**, where each input has a known output. The model learns from this data to predict outputs for new, unseen data.

Examples:

- Spam detection
- House price prediction
- Student performance prediction

Algorithms:

1. Linear Regression
2. Logistic Regression
3. Decision Tree
4. Random Forest
5. Support Vector Machine (SVM)
6. K-Nearest Neighbors (KNN)

B. Unsupervised Learning

Definition: Unsupervised learning uses **unlabeled data**. The model identifies hidden patterns, relationships, or groups in the data without predefined output labels.

Examples:

- Customer segmentation
- Market basket analysis
- Identifying patterns in customer behavior

Algorithms:

1. K-Means Clustering
2. Hierarchical Clustering
3. Principal Component Analysis (PCA)

C. Reinforcement Learning

Definition: Reinforcement learning is a type of ML in which an **agent learns by interacting with an environment**. It receives rewards for desirable actions and penalties (or lower rewards) for undesirable actions, helping it learn a strategy over time.

Examples:

- Game playing
- Robotics
- Autonomous driving research
- Robot navigation

Key Terms:

- **Agent:** The learner or decision-maker.
- **Environment:** The world in which the agent operates.
- **Action:** A decision made by the agent.
- **Reward:** Feedback received after taking an action.

Quick Comparison

Type	**Data Used**	**Main Purpose**
Supervised Learning	Labeled data	Predict outputs
Unsupervised Learning	Unlabeled data	Discover patterns
Reinforcement Learning	Feedback from interactions	Learn the best actions through rewards

Remember:

- Supervised Learning → Learn from labeled examples.
- Unsupervised Learning → Find patterns in unlabeled data.
- Reinforcement Learning → Learn through rewards and penalties.

Machine Learning Notes: Linear Regression

1. What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value by identifying the relationship between an independent variable and a dependent variable.

Examples:

- Predicting house prices based on area.
- Predicting marks based on study hours.
- Predicting product prices based on size.
- Predicting sales based on advertising expenditure.

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2. Linear Regression Equation

The equation of a straight line is:

Y=mX+b

Symbol	Meaning
Y	Dependent variable — the value we predict
X	Independent variable — the input

m	Slope — how much Y changes when X increases by one unit
b	Y-intercept — the predicted value of Y when X=0

Understanding the slope (m)

The slope tells us how much the predicted value of Y changes when X increases by one unit.

- **Positive slope:** Y increases as X increases.
- **Negative slope:** Y decreases as X increases.
- **Zero slope:** Y remains constant as X changes.

Understanding the intercept (b)

The intercept is the predicted value of Y when X=0.

For example, if:

Y=1.5X-2

Then:

- Slope (m) = 1.5
- Intercept (b) = -2

When X=0:

Y=1.5(0)-2=-2

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3. Example: Predicting House Prices

Suppose we have the following dataset:

House Size (X)	House Price (Y)

8	10
10	13
12	16

Here:

- **Independent variable (X):** House size
- **Dependent variable (Y):** House price

Our objective is to find the best-fit line:

$$Y=mX+b$$

To find this line, we calculate the slope and intercept.

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4. Calculating the Mean

The mean is the average of the values.

Mean of X

$$\bar X=\frac{\sum X}{n}$$

$$\bar X=\frac{8+10+12}{3}$$

$$\boxed{\bar X=10}$$

Mean of Y

$$\bar Y=\frac{\sum Y}{n}$$

$$\bar Y=\frac{10+13+16}{3}$$

$$\boxed{\bar Y=13}$$

Where:

- \bar X = Mean of X
- \bar Y = Mean of Y
- n = Number of data points

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5. Calculating Deviations

A deviation represents the difference between an individual value and its mean.

Formulas

Deviation of X:

X-\bar X

Deviation of Y:

Y-\bar Y

Using our dataset:

X	Y	X-\bar X	Y-\bar Y	Product of deviations	Square of X deviation
8	10	-2	-3	6	4
10	13	0	0	0	0
12	16	2	3	6	4
Total		**0**	**0**	**12**	**8**

Important formulas

Product of deviations:

$(X-\bar X)(Y-\bar Y)$

Square of X deviation:

$(X-\bar X)^2$

The totals are:

$\sum (X-\bar X)(Y-\bar Y)=12$

$\sum (X-\bar X)^2=8$

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6. Calculating the Slope (m)

The formula for the slope of the best-fit line is:

$$\boxed{m=\frac{\sum (X-\bar X)(Y-\bar Y)}{\sum (X-\bar X)^2}}$$

Substituting the values:

$$m=\frac{12}{8}$$

Therefore:

$$\boxed{m=1.5}$$

Interpretation: For every one-unit increase in X, the predicted value of Y increases by 1.5 units.

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7. Calculating the Y-Intercept (b)

The formula is:

$$\boxed{b=\bar Y-m\bar X}$$

We know:

- $Y=13$
- $m=1.5$
- $X=10$

Substituting:

$$b=13-(1.5\times10)$$

$$b=13-15$$

$$\boxed{b=-2}$$

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8. Final Linear Regression Equation

We have calculated:

$$m=1.5,\qquad b=-2$$

Using the equation:

$$Y=mX+b$$

Our final equation is:

$$\boxed{Y=1.5X-2}$$

This is the **linear regression equation** for our dataset.

Making a prediction

Suppose $X=15$.

Substitute into the equation:

$$Y=1.5(15)-2$$

$$Y=22.5-2$$

$$\boxed{Y=20.5}$$

The predicted value of Y is **20.5**.

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9. Quick Revision Cheat Sheet

Concept	Formula

Linear regression equation	Y=mX+b
Mean of X	\bar X=\frac{\sum X}{n}
Mean of Y	\bar Y=\frac{\sum Y}{n}
Deviation of X	X-\bar X
Deviation of Y	Y-\bar Y
Product of deviations	(X-\bar X)(Y-\bar Y)
Square of X deviation	(X-\bar X)^2
Slope	m=\frac{\sum(X-\bar X)(Y-\bar Y)}{\sum(X-\bar X)^2}
Y-intercept	b=\bar Y-m\bar X

Remember these key points

1. Linear regression is a **supervised learning** algorithm.
2. It is used to predict continuous numerical values.
3. X is the independent variable, and Y is the dependent variable.
4. The slope (m) determines the direction and rate of change.
5. The intercept (b) gives the predicted value of Y when X is zero.
6. The goal is to find the **best-fit line** that minimizes the sum of squared prediction errors.

One-line summary: Linear regression finds the relationship between input and output variables using a straight-line equation to make predictions.

10. Mean Absolute Error (MAE)

- To predict Student's Marks

Std	Actual Marks	Predicted Marks	Error
A	90	88	2
B	75	80	-5
C	60	55	5
D	95	97	-2
E	70	68	2

Mean Absolute Error (MAE) = (2+5+5+2+2)/5 = 3.2

Std	Actual Marks	Predicted Marks	Error	Error Square
A	90	88	2	2^2 = 4
B	75	80	-5	-5^2 = 25
C	60	55	5	5^2 = 25
D	95	97	-2	-2^2 =4
E	70	68	2	2^2 = 4

Mean Square Error = (4+25+25+4+4) / 5 = 12.4

Squared Error = Square Root of 12.4 = 3.5

LOGISTIC REGRESSION

Class Notes • Sigmoid Function • Standardization • Model Evaluation

These notes are based on the classroom whiteboard examples. They cover the sigmoid function, Logistic Regression calculations, standardization, confusion matrix, accuracy, precision, recall, and a basic Python workflow.

1. What is Logistic Regression?

Logistic Regression is a supervised machine learning algorithm primarily used for classification. It estimates the probability that an input belongs to a particular class.

Example: predicting whether a student passes or fails based on exam preparation.

Example input	Predicted class
2 hours of study	Fail (0)
3 hours of study	Fail (0)
5 hours of study	Pass (1)
7 hours of study	Pass (1)

Other applications include spam detection, disease prediction, and customer churn prediction.

Important: Despite its name, Logistic Regression is primarily used for classification, whereas Linear Regression is generally used to predict continuous numerical values.

2. The Sigmoid Function

The sigmoid function converts a model's linear output into a probability between 0 and 1.

$$\sigma(z) = 1 / (1 + e^{-z})$$

For a single input feature:

$$z = b_0 + b_1 x$$

Where:

- $\sigma(z)$ = predicted probability
- z = linear combination of the input features and model parameters
- e = Euler's number, approximately 2.718
- b_0 = intercept (bias)
- b_1 = coefficient of the input feature
- x = input feature

Complete equation:

$$P(y = 1 \mid x) = 1 / (1 + e^{-(b_0 + b_1 x)})$$

How the sigmoid function behaves

z	Approximate probability
-5	0.0067
-2	0.1192

0	0.5000
2	0.8808
5	0.9933

Classification threshold

A common default threshold is 0.5:

- If probability ≥ 0.5, predict class 1.
- If probability < 0.5, predict class 0.

Example: probability 0.81 → Pass (1); probability 0.35 → Fail (0). The threshold can be changed for different applications.

3. Worked Logistic Regression Example

Given $b_0 = -1.5$, $b_1 = 0.6$, and $x = 5$, predict whether the student passes or fails.

Step 1: Calculate z

$z = b_0 + b_1x = -1.5 + (0.6 \times 5) = -1.5 + 3 = 1.5$

Step 2: Apply the sigmoid function

$P = 1 / (1 + e^{-1.5}) \approx 1 / (1 + 0.2231) \approx 0.8176$

Step 3: Make the prediction

Since 0.8176 ≥ 0.5, the predicted class is Pass (Class 1).

Interpretation: The model assigns an estimated 81.76% probability to the Pass class. This is the model's predicted probability, not necessarily the student's actual chance of passing.

4. Standardization

Standardization transforms numerical features so that they have a mean of approximately 0 and a standard deviation of approximately 1.

$$Z = (X - \mu) / \sigma$$

Where X = original value, μ = feature mean, and σ = feature standard deviation.

Example

If X = 80, μ = 60, and σ = 10:

$$Z = (80 - 60) / 10 = 2$$

The value is two standard deviations above the mean.

Why standardization is useful

Features such as age, salary, and study hours use different scales. Standardization puts them on comparable scales and can help the numerical optimization used to train Logistic Regression.

Python example

```
from sklearn.preprocessing import StandardScaler

 scaler = StandardScaler()

 X_train_scaled = scaler.fit_transform(X_train)
 X_test_scaled = scaler.transform(X_test)
```

Important: fit the scaler on training data only. Use transform() on test data to prevent data leakage.

5. Confusion Matrix

A confusion matrix compares predicted classes with actual classes.

For the student example, positive = Pass and negative = Fail.

Actual / Predicted	Predicted Negative (Fail)	Predicted Positive (Pass)
Actual Negative (Fail)	True Negative (TN)	False Positive (FP)
Actual Positive (Pass)	False Negative (FN)	True Positive (TP)

The four outcomes

True Positive (TP): Actual positive and predicted positive. Example: a student passes and the model predicts Pass.

True Negative (TN): Actual negative and predicted negative. Example: a student fails and the model predicts Fail.

False Positive (FP): Actual negative but predicted positive. Example: a student fails but the model predicts Pass. Also called a Type I error.

False Negative (FN): Actual positive but predicted negative. Example: a student passes but the model predicts Fail. Also called a Type II error.

Worked confusion matrix

Suppose TP = 3, TN = 4, FP = 1, and FN = 2.

Actual / Predicted	Predicted Fail	Predicted Pass
Actual Fail	TN = 4	FP = 1
Actual Pass	FN = 2	TP = 3

Total predictions = TP + TN + FP + FN = 3 + 4 + 1 + 2 = 10.

6. Accuracy

Accuracy is the proportion of all predictions that are correct.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Using TP = 3, TN = 4, FP = 1, and FN = 2:

Accuracy = (3 + 4) / 10 = 7 / 10 = 70%.

Interpretation: 70% of the model's predictions are correct.

Limitation of accuracy

If 95 out of 100 students fail, a model that predicts Fail for everyone achieves 95% accuracy but identifies none of the students who pass. Accuracy alone can therefore be misleading for imbalanced datasets.

7. Precision

Precision measures the proportion of predicted positive cases that are actually positive.

Precision = TP / (TP + FP)

Using TP = 3 and FP = 1:

Precision = 3 / (3 + 1) = 3 / 4 = 75%.

Interpretation: The model predicted Pass for 4 students, and 3 actually passed.

Precision is useful when false positive predictions are costly. In spam detection, it helps measure how many emails marked as spam really are spam.

8. Recall

Recall measures the proportion of actual positive cases that the model correctly identifies.

Recall = TP / (TP + FN)

Using TP = 3 and FN = 2:

Recall = 3 / (3 + 2) = 3 / 5 = 60%.

Interpretation: 5 students actually passed, and the model correctly identified 3 of them.

Recall is important when missing positive cases is costly. For example, high recall is often desired in disease screening.

9. Accuracy vs Precision vs Recall

Metric	Formula	Question answered
Accuracy	(TP + TN) / Total	How many total predictions were correct?
Precision	TP / (TP + FP)	Of predicted positives, how many were actually positive?
Recall	TP / (TP + FN)	Of actual positives, how many were correctly identified?

The **F1 Score** is the harmonic mean of Precision and Recall. It balances both metrics.

\boxed{F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}}

Where:

- **Precision** = \frac{TP}{TP+FP}
- **Recall** = \frac{TP}{TP+FN}

Example (using your classroom values)

Given:

- TP = 3
- FP = 1
- FN = 2

10. Calculate all three metrics

Metric	Calculation	Result
Accuracy	(3 + 4) / 10 × 100	70%
Precision	3 / (3 + 1) × 100	75%
Recall	3 / (3 + 2) × 100	60%

These metrics describe different aspects of the same model. A model can have high precision but relatively low recall, or vice versa.